RDG-P6 · PRÁCTICA 02

Gapped text

Hoja 02 · Divulgación: relato y explicación · dificultad 2 de 4

7 huecos (37–43) · 8 párrafos (A–H), uno sobrante · 2 puntos por hueco · tiempo orientativo: 18 minutos · sin consultar la teoría

You are going to read an extract from a magazine article about a discovery in astronomy. Seven paragraphs have been removed from the extract. Choose from the paragraphs **A–H** the one which fits each gap (37–43). There is one extra paragraph which you do not need to use. Mark your answers on the answer sheet.

A bit of scruff

In 1967 a research student noticed a faint, recurring mark in miles of chart paper. Explaining it would take astronomers into some of the strangest territory in physics.

In the summer of 1967 Jocelyn Bell, a 24-year-old research student at Cambridge, had every reason to feel that the hardest part of her doctorate was behind her. For two years she had helped to build a radio telescope in a field outside the city: more than a thousand wooden posts strung with wire, spread over some four and a half acres. Its purpose was to find quasars, the brilliant cores of distant galaxies, which were then among the most fashionable objects in astronomy. In July the telescope was switched on, and her real work began.

37 _____

One of them kept coming back. It occupied less than half an inch of a trace that ran for hundreds of feet, and she had seen it before, from the same patch of sky. Bell had a name for it, ‘a bit of scruff’, and might reasonably have left it at that. What made it impossible to dismiss was its timing: each day it appeared about four minutes earlier than the day before.

38 _____

Even so, her supervisor, Antony Hewish, was inclined to put it down to man-made interference. In late November Bell was in the field with a high-speed recorder, which spread the signal out so that its structure could be seen, when the scruff appeared again. It was not a smear at all but a string of sharp pulses, separated by intervals of a little over one and a third seconds. Nothing in nature that astronomers knew of kept time like that.

39 _____

Bell herself found the idea more irritating than thrilling: she had a thesis to finish, and the last thing she needed was an alien civilisation choosing her aerial to signal through. Just before Christmas, working late, she found a second patch of scruff in a quite different part of the sky, pulsing at a slightly different rate. Two sets of little green men, transmitting on the same frequency from different corners of the galaxy, seemed a great deal less likely than one.

40 _____

Such a collapse also sets the remnant spinning furiously, for the same reason that a skater turns faster as she draws in her arms. The neutron star’s magnetic field channels radiation into two narrow beams, and because the field is tilted relative to the axis of rotation, the beams sweep round the sky like the light from a lighthouse. A telescope that lies in their path receives a pulse once every turn.

41 _____

Little of this was understood in February 1968, when the discovery was announced in the journal Nature, and the press seized on the story. Hewish was questioned about the astrophysics, and Bell about almost everything else. She was asked how many boyfriends she had and whether she was taller than Princess Margaret, and photographers had her run along a bank waving her arms, to look as if she had just made a discovery.

42 _____

Her own verdict has been notably generous. She has argued that it would cheapen the prizes to award them to research students except in quite exceptional cases, and that hers was not one of them. A supervisor, she has pointed out, bears the responsibility when a project fails, so it seems only fair that he should share in its successes. Not everyone has been persuaded by that reasoning, and the episode is still cited whenever the habits of scientific prizes are discussed.

43 _____

The scholarships support students at exactly the stage she had reached in 1967, when a career consists largely of long hours spent on unglamorous data. It is a fitting legacy for a discovery that began not with a flash of insight but with a refusal to ignore a mark that took up less than half an inch of chart.

A Six years later the discovery was recognised with the Nobel Prize in Physics. It went to Hewish, jointly with his Cambridge colleague Martin Ryle, for their pioneering work in radio astronomy. Bell, by then married and known as Jocelyn Bell Burnell, was not included, and the astronomer Fred Hoyle was among those who said publicly that she should have been.

B If nature could not account for it, perhaps something else could. Half-seriously, the team wondered whether they had picked up a signal from another civilisation, and the source was labelled LGM-1, for 'little green men'. Nobody quite believed it, but nobody could rule it out either, and for several weeks the group said nothing outside the department while they looked for a duller explanation.

C Quasars twinkle for much the same reason that stars do. Their radio waves pass through the thin stream of charged particles flowing outwards from the Sun, which scatters them and makes a sufficiently compact source appear to flicker. The telescope had been designed to exploit precisely this effect, since only the most compact sources twinkle at all.

D The telescope produced nearly a hundred feet of chart paper a day, and there were no computers to read it. Every inch passed through Bell's hands. Within weeks she could tell the flickering trace of a quasar from the interference of a passing car or a badly suppressed thermostat, and she noted each one in the margin. A few squiggles, however, fitted neither category.

E That is why the pulses are so regular. A body this dense cannot speed up or slow down quickly, and the most stable pulsars keep time with an accuracy to rival the best atomic clocks. Astronomers now use networks of them as an instrument the size of the galaxy: in 2023 several teams reported that the timing of dozens of pulsars bore the faint imprint of gravitational waves.

F She was, in any case, to have the last word. In 2018 she received a Special Breakthrough Prize in Fundamental Physics, worth three million dollars, and gave the entire sum away to fund doctoral scholarships for students from groups under-represented in physics, among them women, refugees and ethnic minorities.

G That detail is the signature of the stars. The Earth turns once relative to the Sun every twenty-four hours, but once relative to the distant stars about four minutes sooner, so anything fixed among the stars rises a little earlier each day. Interference from cars and radios keeps human hours. Whatever was producing the scruff was keeping the time of the sky.

H What Bell had found was a new kind of star, or rather the remains of one. When a star much more massive than the Sun runs out of fuel, its core can collapse in a fraction of a second, crushing its matter until it consists mostly of neutrons. The result, a neutron star, packs more than the mass of the Sun into a sphere the size of a city.

Hoja de respuestas

Marca una sola letra por hueco; cada letra se usa una vez como máximo. Al corregir, 2 puntos por acierto; los errores no restan. Anota también el tiempo empleado: el objetivo en el examen son unos 18 minutos.

Nº	A	B	C	D	E	F	G	H	Puntos
37	A	B	C	D	E	F	G	H	/ 2
38	A	B	C	D	E	F	G	H	/ 2
39	A	B	C	D	E	F	G	H	/ 2
40	A	B	C	D	E	F	G	H	/ 2
41	A	B	C	D	E	F	G	H	/ 2
42	A	B	C	D	E	F	G	H	/ 2
43	A	B	C	D	E	F	G	H	/ 2
							Total		**/ 14**

Párrafo sobrante: _____ (no puntúa; sirve para comprobar). Tiempo empleado: _____ minutos.

SOLUCIONES

2 puntos por acierto. Cada solución explica qué une el párrafo con lo de antes (←) y con lo de después (→), con citas de los dos lados; cuando hay un párrafo tentador, por qué falla; y, al final, por qué el sobrante no entra en ningún hueco. Los números de sección remiten a RDG-P6_Teoria_v1.

Clave rápida

Nº	Clave	Empieza	Teoría
37	**D**	The telescope produced nearly a hundred feet…	§4.1 · §4.2
38	**G**	That detail is the signature of the…	§4.1 · §3.3 · §4.3
39	**B**	If nature could not account for it…	§4.1 · §4.4
40	**H**	What Bell had found was a new…	§3.3 · §4.1 · §4.4
41	**E**	That is why the pulses are so…	§4.3 · §3.3
42	**A**	Six years later the discovery was recognised…	§3.1 · §4.1
43	**F**	She was, in any case, to have…	§4.5 · §4.1
—	**C**	Quasars twinkle for much the same reason… *(sobrante)*	§5.1 · §3.3

Explicación por hueco

37 · D · *The telescope produced nearly a hundred feet of chart…*

← Con lo anterior. El primer párrafo termina con el telescopio en marcha (In July the telescope was switched on). D lo retoma con artículo definido y cuenta en qué consistía el trabajo: The telescope produced nearly a hundred feet of chart paper a day.

→ Con lo siguiente. D termina con unos garabatos que no encajan en ninguna categoría: A few squiggles, however, fitted neither category. El párrafo siguiente escoge uno de ellos con un pronombre: One of them kept coming back. Sin un plural delante, One of them no tiene a qué referirse.

Tentador: C. Encaja por detrás, porque explica cómo se detectan los quásares que menciona el primer párrafo (Its purpose was to find quasars). Pero C no contiene ningún grupo del que One of them pueda escoger uno, ni nada que se resista a ser clasificado.

Teoría: §4.1 Pistas hacia atrás · §4.2 Pistas hacia delante

38 · G · *That detail is the signature of the stars.*

← Con lo anterior. El párrafo anterior termina con un dato preciso: each day it appeared about four minutes earlier than the day before. G lo recoge con un sustantivo resumen, That detail, y cambia al presente para explicarlo: es el paso del relato a la explicación (§3.3).

→ Con lo siguiente. G concluye que la señal sigue el tiempo de las estrellas, no el de las personas: keeping the time of the sky. El párrafo siguiente empieza con Even so: admite ese indicio y presenta una reacción contraria: Hewish sigue pensando en una interferencia humana (§4.3).

Tentador: H. What Bell had found parece resolver el enigma, pero en este punto la historia no ha resuelto nada: después viene un supervisor que sospecha de una interferencia (Even so, her supervisor) y todavía faltan los pulsos y la segunda señal. Además, H es lo único que puede preparar Such a collapse, que está tres párrafos más abajo.

Teoría: §4.1 Pistas hacia atrás · §3.3 Textos mixtos · §4.3 Conectores y adverbios de frase

39 · B · *If nature could not account for it, perhaps something…*

← Con lo anterior. El párrafo anterior acaba con una regularidad que nada natural explica: Nothing in nature that astronomers knew of kept time like that. B parte de esa conclusión y da el paso siguiente: If nature could not account for it, perhaps something else could. El pronombre it es la señal que marca el tiempo, y nature retoma el Nothing in nature de la frase anterior.

→ Con lo siguiente. B introduce la hipótesis extraterrestre (little green men). El párrafo siguiente la presupone dos veces: Bell herself found the idea more irritating than thrilling y Two sets of little green men.

Tentador: E. That is why the pulses are so regular parece encajar tras los pulsos regulares, pero That is why exige una explicación previa, y la del faro todavía no ha aparecido. Además, the idea del párrafo siguiente se quedaría sin antecedente.

Teoría: §4.1 Pistas hacia atrás · §4.4 Cadenas léxicas

40 · H · *What Bell had found was a new kind of…*

← Con lo anterior. El párrafo anterior descarta la hipótesis extraterrestre (seemed a great deal less likely than one). H pasa del relato a la explicación con un puente típico: What Bell had found was a new kind of star (§3.3).

→ Con lo siguiente. H describe el colapso del núcleo (its core can collapse in a fraction of a second) y lo que queda de la estrella (the remains of one). El párrafo siguiente recoge las dos cosas: Such a collapse also sets the remnant spinning furiously.

Teoría: §3.3 Textos mixtos · §4.1 Pistas hacia atrás · §4.4 Cadenas léxicas

41 · E · *That is why the pulses are so regular.*

← Con lo anterior. El párrafo anterior termina con el mecanismo de los pulsos: receives a pulse once every turn. E lo convierte en causa: That is why the pulses are so regular.

→ Con lo siguiente. E llega hasta 2023 (in 2023 several teams reported), y el párrafo siguiente vuelve al relato con una fecha y con un resumen de toda la explicación: Little of this was understood in February 1968. Es el puente de vuelta de §3.3.

Teoría: §4.3 Conectores y adverbios de frase · §3.3 Textos mixtos

42 · A · *Six years later the discovery was recognised with the…*

← Con lo anterior. El párrafo anterior cuenta el anuncio del descubrimiento en febrero de 1968, y A sigue la cronología: Six years later. El contraste entre Hewish, a quien preguntan por la ciencia (Hewish was questioned about the astrophysics), y Bell, a quien preguntan por su vida privada, prepara la exclusión que cuenta A.

→ Con lo siguiente. A termina con la polémica: Bell quedó fuera, y Fred Hoyle was among those who said publicly that she should have been. El párrafo siguiente da la opinión de la propia Bell sobre esa exclusión: Her own verdict. El adjetivo own la contrapone a la de Hoyle.

Tentador: F. She was, in any case, to have the last word podría cerrar la escena de la prensa, pero el párrafo siguiente (Her own verdict) necesita una decisión que juzgar, y F no la contiene: el Nobel solo aparece en A.

Teoría: §3.1 Hilo narrativo · §4.1 Pistas hacia atrás

43 · F · *She was, in any case, to have the last…*

← Con lo anterior. El párrafo anterior deja la polémica abierta: Not everyone has been persuaded by that reasoning. F la cierra con un adverbial concesivo y una anticipación: She was, in any case, to have the last word. La forma was to have anuncia el salto hasta 2018 (§4.5).

→ Con lo siguiente. F termina con el destino del dinero (to fund doctoral scholarships), y el último párrafo lo presupone con artículo definido: The scholarships support students.

Teoría: §4.5 Tiempo verbal y voz · §4.1 Pistas hacia atrás

El párrafo sobrante

C · *Quasars twinkle for much the same reason that stars…*

- **Hueco 37.** Encaje por un solo lado. Por detrás funciona, porque explica para qué servía el telescopio (Its purpose was to find quasars); pero el párrafo siguiente empieza con One of them kept coming back, y C no contiene ningún grupo de señales sin clasificar.
- **Hueco 38.** C no explica el dato que cierra el párrafo anterior (about four minutes earlier), y Even so necesita delante un indicio de que la señal no es una interferencia humana.
- **Huecos 39 y 40.** Tras los pulsos y tras la segunda señal, C cambia de tema sin puente (§3.4), y lo que sigue a cada hueco presupone algo que C no contiene: the idea y little green men en el 39, Such a collapse en el 40.
- **Hueco 41.** Podría seguir a la explicación del faro, pero vuelve a los quásares sin preparación, y Little of this was understood in February 1968 chocaría con C: el centelleo se conocía bien, puesto que el telescopio se había construido para aprovecharlo (The telescope had been designed to exploit precisely this effect).
- **Huecos 42 y 43.** En la parte final el texto trata de la prensa, el Nobel y la reacción de Bell; C no tiene relación con ese hilo, y lo que sigue a cada hueco (Her own verdict, The scholarships) presupone algo que C no menciona.

Teoría: §5.1 Cómo se fabrica el sobrante · §3.3 Textos mixtos